25900223 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZIM102 - MOLECULES OF LIFE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. Glucose is a type of ……..that serve as primary source of energy. [K] / [CO1]
2. The two ends of a polypeptide chain are known as.....terminal and carboxy terminal. [U] / [CO3]
3. Fatty acids composed of carboxylic acid head group and a long .....chain [U] / [CO4]
4. Glycolipid play a crucial role in.....recognition and cell signalling. [U] / [CO5]
5. .....is widely used as a supporting agent to culture bacterial colonies. [K] / [CO1]
6. ..... is a digestive enzyme that plays a crucial role in the digestion of Maltose. [U] / [CO1]
7. Globular proteins are.....in shape. [K] / [CO2]
8. The number of grams of iodine absorbed by 100 gm of fat or oil is called..... [U] / [CO4]

9. Hemoglobin is a …….. protein found in red blood cells. [K] / [CO3]

10. The angle of rotation in peptide bond is known as.....plot. [U] / [CO2]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Name the chemical formula of mannose and galactose and how it differ from each other. [K] / [CO1]

12. Discuss the characteristics of globulin protein. [U] / [CO2]

13. Give an example of a nucleotide that serve as an energy carrier in cells. [K] / [CO5]

14. What are acidic amino acids? [U] / [CO2]

15. Outline clinical significance of PUFA. [K] / [CO5]

16. Explain the compostion of Bacterial Cell wall. [U] / [CO1]

17. Discuss the role of contractile protein in human body. [K] / [CO2]

18. Discuss the classification of fatty acid based on chain length and number of carbon atom.. [U] / [CO4]

19. Write short note on homopolysaccharides. [K] / [CO1]

20. How did Chargaff's rule contribute to the discovery of the DNA double helix. [U] / [CO5]

21. Name and explain one specific test to detect fructose. [K] / [CO1]

22. Discuss the reaction and enzymes involved in transamination of amino acids. [U] / [CO2]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Explain the role of phospholipids and glycolipids in cell memberane. [K] / [CO5]

24. Differniate between epimer and anomer. [U] / [CO1]

25. Differentiate Purely ketogenic amino acids from glucogenic amino acids. [K] / [CO2]

26. Write an essay on different types of carbohydrates based on structure. [U] / [CO1]

27. Explain the relationship between primary ,secondary ,tertiary and quaternary structure of proteins. [K] / [CO3]

28. Explain the role of nucleic acids in forensic science and medical diagnostics. [U] / [CO5]

29. Differentiate between starch and Glycogen. [K] / [CO1]

[4x5 = 20]